

Subjectivity-Intersection Braid Quantum Gravity

A Constructive Refutation of Objectivist Primacy from a Source-Derived Lorentz Carrier and a Conditional Einstein Limit

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Claim status. This is a candidate theory, not completed or empirically confirmed quantum gravity. All v0.6.1 finite geometry, action, locality, and code-audit boundaries are retained. Post-v0.6.1 exact results close MMR2 routing inside the declared fixed event-conditioned model and derive the relative factor 3/5 without manual setting. The actual Braid event transition produces a full-rank four-direction moment; combined with the independently source-selected causal grading it gives an exact Lorentz-signature carrier with a unique normalization in a stated naturality class. Full source-to-continuum convergence remains unproved, and the Source-Dynamical Parent Completion (SDPC)—action-level causal grading, physical metric/coframe pullback, nontrivial physical-section dynamics, and the Ward identity from one parent action—is not yet derived from Braid alone. Hence $G_{\mu\nu} = (3/5)T_{\mu\nu}$ remains a conditional continuum theorem under the retained continuum hypothesis and SDPC.

Abstract

We formulate *Subjectivity-Intersection Braid Quantum Gravity* as a candidate theory in which quantum kinematics, spacetime geometry, and gravitational response arise from one standpoint-bearing pointed-braid source rather than being postulated as independent primitives. Here subjectivity denotes a mathematically or physically located standpoint of access and transport, not consciousness. Distinct source standpoints remain nonidentical, while their Braid-mediated intersection generates a covariant structure shared across changes of presentation.

The source is quantum in the algebraic sense: a noncommutative matrix-algebra tower carries states, unitary pointed-braid transport, and Yang–Baxter-coherent evolution. Its observer quotient is $Q_{\text{obs}} = V_4$. Requiring a real readout to contain the common order unit, separate the four observer states, and respect the full intrinsic affine action forces the complete function carrier. It splits canonically as one constant direction plus a three-dimensional mean-zero representation. Thus dimension four and the $1 + 3$ type are outputs of a stated observer-completeness principle rather than requested target dimensions. Under Observer Uniform Cauchy Closure, the rational carrier completes without new directions to $W = \mathbb{R}^{V_4}$, whose affine space $\text{Aff}(W)$ is a smooth four-dimensional carrier.

Exact finite certificates further generate a nonzero covariant symmetric tensor carrier, source-native coframes and holonomies, finite metric-affine gravity equations with torsion and nonmetricity, graph-local propagation, a source-local GKSL evolution, nonzero connection curvature, first-order curvature and defect actions, and discrete Noether identities. These constructions regenerate through all declared finite refinement stages without stage-dependent fitting. The one-dimensional source path is not labelled as three-dimensional physical space: it indexes operator locality, while the source-derived $1 + 3$ representation supplies the local event type.

We then separate pointwise arrival from variational derivation. In v0.5 the four historical bridge laws were compressed to CGR and MMR. The spectral marker yields rank-one event atoms; the actual crossing passes an exact 25-event and set-theoretic Yang–Baxter check. This discharges a finite *event-algebra* component of CGR. A separate smooth cylinder completion is mathematically constructible, but existing code samples chosen smooth fields into finite edge data; it does not prove that the actual finite Braid field sequence converges back to that completion. Formal CGR as a whole therefore remains open at the source-to-continuum arrow, as does natural-spacetime identification. On the selected smooth cylinder, the original G_1 Gibbs functional has a local doubled matter-action construction (MMR1); the ten finite response operators lie in a source-fixed event/energy operator span.

Version 0.6.1 correctly recorded that the then-tested relational mark link (RML) did not by itself select MMR2. The post-v0.6.1 fixed-packet audit changes that dependency without rewriting the older result: among the complete five-element partition-retraction class generated by the three marked source atoms, only the full corner preserves the already fixed eleven-operator event-conditioned packet. Thus MMR2 is closed inside that declared fixed model, and the normalized source cup yields 3/5 without manual insertion. This does not classify arbitrary completely positive instruments, arbitrary subalgebras, natural signed-sector selection, or empirical matter.

The actual involutive Braid event transition also supplies a permutation-covariant positive rank-four moment Q_4 . Four certified source-null links reconstruct a unique Lorentz metric, and the independently source-selected causal reflection J_{ray} gives $K_{\text{evt}} = J_{\text{ray}}Q_4$ with exact signature $(3+, 1-)$. Its positive self-adjoint S_4 -equivariant normalization to the inverse source-null Gram tensor is unique. No Wick rotation, manual sign, or continuous coefficient fit enters this carrier result. The retained positive relative-entropy edge action does not yet select the signed grading; orientation-only and fixed closed-time-path branch routes fail exactly. These results close the algebraic Lorentz carrier, not the parent action.

The remaining matter/stress requirement is compressed into the post-v0.6.1 *Source-Dynamical Parent Completion* (SDPC): one local parent action must reproduce the fixed source first jet, select the physical metric/coframe pullback and causal principal symbol, contain nontrivial physical-section dynamics, and yield the Ward identity from its diagonal covariance. Together with the still-explicit source-to-continuum convergence and first-variation hypothesis, SDPC gives the infrared action

$$S_{\text{IR}} = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \psi]$$

yields, in the long-wavelength, low-curvature, matter-hypermomentum-free sector,

$$G_{\mu\nu}(g) = \kappa_B T_{\mu\nu}, \quad \kappa_B = \frac{3}{5},$$

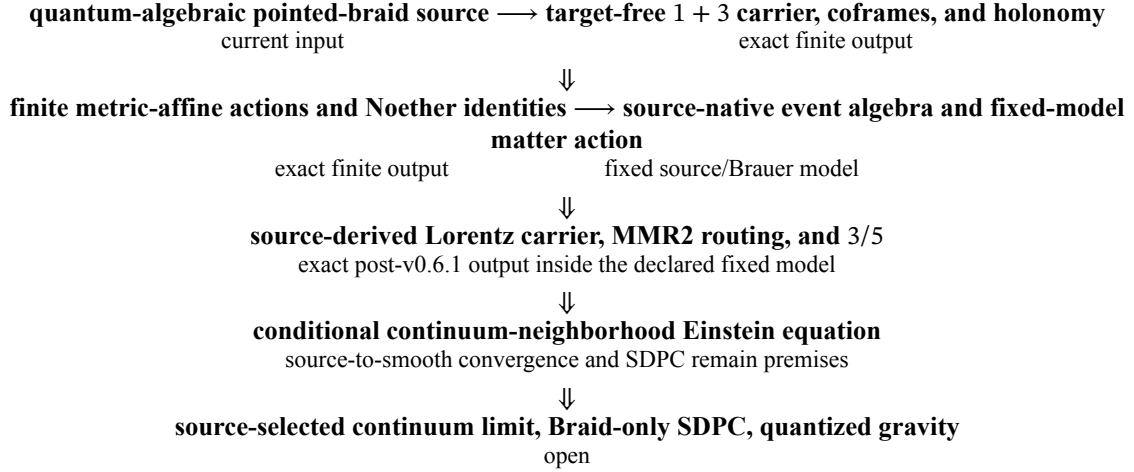
as a conditional Euler–Lagrange field equation on a continuous spacetime neighborhood rather than as a fit at one point.

The formal countermodel continues to challenge the precisely defined necessity thesis of *objectivist primacy*: a shared tensor carrier, its 1 + 3 type, and a Lorentz-signature event carrier need not be primitive in the declared model. It does not refute scientific realism in general or demonstrate that the model describes nature. Source-to-continuum convergence, Braid-only SDPC, continuum gravitational quantization, ultraviolet completion, and empirical gravitational identification remain open. Subjectivity denotes standpoint, not consciousness.

Keywords: Subjectivity-Intersection Braid Quantum Gravity; quantum gravity candidate; pointed braid; generated objectivity; objectivist primacy; continuum Einstein equation; metric-affine gravity; emergent spacetime; Creation Principle; Yang–Baxter coherence.

1 The Braid-first question

The proposal is intentionally ambitious: begin from one standpoint-bearing pointed-braid source and ask how much of the architecture normally assigned separately to quantum theory, spacetime, and gravity can be recovered as outputs of one generative law. The present evidence establishes the middle finite part of that programme, not its endpoints. Its logical status is summarized by



The research wager is that the remaining continuum, parent-action, and physical-identification conditions can ultimately be supplied by the same source law rather than attached afterward. Since v0.5, a finite event algebra has become source-native, a doubled matter action has been defined on a selected smooth cylinder, MMR2 has been closed inside the fixed event-conditioned model, the relative factor 3/5 has been routed without manual setting, and a source-derived Lorentz event carrier has been obtained. None of these results silently proves actual source-to-continuum convergence or the full SDPC parent action.

Modern physics succeeds by extracting quantities that do not depend on arbitrary coordinates, gauges, or laboratory presentations. The methodological value of objectivity is not in dispute here. The disputed step is ontological: must the invariant structure be there first, independently of every standpoint, with perspectives entering only as partial descriptions of it? Call that stronger commitment *objectivist primacy*. It is stronger than scientific reproducibility, stronger than observer agreement, and stronger than covariance.

The distinction matters because invariance alone does not specify the order of construction. One may begin with an objective object and derive its appearances. Alternatively, one may begin with distinct standpoints and derive a public invariant by quotienting only their representational redundancy. Both routes can end with an invariant. They disagree about what is primitive.

This paper develops the second route as the proven finite core and conditional continuum extension of Subjectivity-Intersection Braid Quantum Gravity. The mathematical work is motivated by Subjectivity Intersection Mathematics and its pointed-unitary-braid realization [22, 20]. Its first aim is not to replace the predictions of established physics. It is to establish that the same source can generate classical objective structure and finite gravity dynamics, and can support a conditional continuum Einstein law, so that the primacy of objectivity is not forced by exactness, covariance, reproducibility, or stable shared structure.

The proposal is close enough to relational quantum mechanics, quantum reference frames, QBism, perspectival realism, and structural realism that careful separation is essential. Relational quantum mechanics denies an observer-independent quantum state and treats facts as relational [15]. Quantum-reference-frame formalisms construct transformations among descriptions associated with physical systems and generalize covariance without an external absolute frame [6, 17]. QBism treats the quantum state as an agent's personal probabilistic assignment rather than an objective property [5]. Perspectival realism studies reliable scientific knowledge across situated perspectives [12, 13]. Structural realism and invariantism identify objective content with transformation-invariant structure [19, 11]. The present proposal shares relational and perspectival motivations, but makes a different technical claim: an objective tensor carrier is an *output* of the maintained distinction and intersection of two source standpoints, and the same output is regenerated at every finite stage of a nonterminal braid tower.

2 Three claims that must not be conflated

2.1 Subjectivity is not consciousness

In this paper, a *subjectivity* is a located mode of access, transport, or presentation that cannot simply be deleted while preserving the construction. It may be represented by a physical reference system, a source prefix, or a typed algebraic standpoint. Nothing in the mathematics attributes sensation, awareness, phenomenal experience, intention, or agency to a braid, a photon, a clock, or an algebra. In particular, *Relative Subjectivity is not an individual subject*: it is not defined by a person, organism, brain, mind, or conscious observer. It names the non-personal relative mode associated, in the proposed physical interpretation, with time, clocking, ordering, and comparison. Human and other biological subjects may instantiate highly organized forms of relative standpoint, but they do not constitute or originate Relative Subjectivity.

This usage is ontological rather than psychological. It says that a physical description is made from somewhere in the structure, and that the difference among such “somewheres” can be constitutive rather than merely epistemic. On the proposed ontology, Relative Subjectivity therefore belongs to physical reality prior to life: clock-like relative ordering and standpoint difference do not begin with humanity, and would obtain in a prebiotic universe. This temporal priority is an ontological interpretation of the physical identification, not a theorem of the present Braid certificates. A separate bridge axiom and separate evidence would be required to connect this notion to consciousness.

2.2 The formal generated-objectivity thesis

Let A and B denote distinct typed source standpoints and let C denote the structure produced from their intersection after allowable presentation dependence has been removed.

Definition 2.1 (Generated objectivity). A carrier $C(A, B)$ exhibits *generated objectivity* if:

1. it is invariant under internal changes of presentation of A and B ;
2. it obeys the appropriate covariance law under a change of ambient description;
3. exchanging the two standpoints does not change the joint carrier;
4. collapsing either standpoint changes the carrier;
5. one source-native law regenerates the carrier through the declared refinement stages without stage-dependent fitting.

Conditions 1–3 provide public stability. Condition 4 prevents “objectivity” from being obtained by deleting the very distinction that was claimed to generate it. Condition 5 makes the carrier a repeated result of a law rather than a one-time finite artifact.

2.3 The ontological Creation Principle

The author’s pre-spacetime interpretation uses the following names:

$$\begin{aligned} A &= \text{light} / \text{Absolute Subjectivity}, & B &= \text{time-clock} / \text{Relative Subjectivity}, \\ C &= \text{space} / \text{Intersection Subjectivity}. \end{aligned}$$

Here B does not denote a human observer or an individual perspective. It denotes an impersonal relative pole whose proposed physical realization is time or clock relation. Accordingly, this ontology places Relative Subjectivity before persons, organisms, and consciousness; its existence is not contingent on the later appearance of humanity. The labels are not premises of the tensor calculation. They constitute an interpretation map from the formal roles to a proposed ontology.

Hypothesis 2.2 (Creation Principle). Objective physical structure is not an independently completed container in which subjectivities later appear. Distinct subjectivities remain nonidentical. Through their intersection they generate a shareable structure C . Re-entry at later stages regenerates rather than merely transports that structure.

The formal result below supports this hypothesis by providing an exact model of its structural grammar. It does not establish the interpretation map to light, time, and physical space. The map becomes a physical theory only when it supplies dynamics, scale, operational readouts, and empirical discrimination.

2.4 The exact target of the counterexample

To avoid attacking a vague “objectivism,” define the target precisely.

Definition 2.3 (Objectivist Primacy Thesis, OP). Every mathematically admissible construction of a stable, intersubjectively shareable, transformation-respecting physical carrier must take that objective carrier, or an equivalent perspective-neutral carrier, as primitive. Distinct standpoints can describe it but cannot generate it.

OP is a universal necessity claim. It is not identical to scientific realism, mind-independent reality, or ordinary experimental objectivity, or the claim that invariant quantities are especially informative. One countermodel meeting the advertised conditions is sufficient to refute OP as stated. It is not sufficient to prove that the countermodel is realized in nature.

3 A source-first pointed-braid setting

3.1 Finite source, nonterminal law

Let $\mathcal{A}_n = M_{5n}(\mathbb{C})$ denote the finite matrix algebra at stage n , with the refinement

$$j_n : \mathcal{A}_n \longrightarrow \mathcal{A}_{n+1}, \quad j_n(X) = X \otimes I_5. \quad (1)$$

Every stage is finite. The sequence has no terminal stage. This distinction is important: a nonterminal tower is not, by itself, a continuum, an infinite-mode quantum field, or physical spacetime. It supplies an exact setting in which one law can be tested at arbitrarily many finite stages.

The source is pointed and braided: it carries a distinguished state and typed unitary braid transports satisfying the declared Yang–Baxter, insertion, and composition laws. The all-orders endpoint-consistency theorem and the positive bilateral response operators were established previously [20]. The present paper uses that source theorem but does not infer physical subjectivity from it.

The computational realization contains eight signed source sectors. Each sector supplies the primitive source packet

$$Q_2 = (R, H, F, P), \quad (2)$$

where the symbols denote the registered relation and source fields, not independently fitted target geometry. Three selected signed-braid generators act as commuting involutions on the certified orbit. From one four-state source slice they generate the opposite slice. Exact checks recover all 12 declared source updates, signed relations, clock fields, and associated source covectors; the identity-map comparator fails in all 12 cases.

3.2 Source-native geometry before the present carrier

The relevant prior source chain is

$$(R, H, F, P) \longrightarrow \text{signed-braid source orbit} \longrightarrow \text{affine split-frame geometry} \longrightarrow \text{coframe and holonomy.} \quad (3)$$

The end-to-end constructor uses no stored future geometry. It reconstructs the connection data, split-frame defect, current term, affine edge, future spatial edges, and future faces from the predicted source relation. In the certified fixed orbit it recovers 12/12 source-slice updates, 24/24 affine edges, 24/24 future spatial edges, and 6/6 future faces exactly. Removing either of two required source sectors breaks the corresponding controls.

This result removes one important circularity: the geometry used below is not inserted as a target answer and then redescribed as braid-generated. It is the output of the registered finite source constructor. The constructor remains global on the fixed finite orbit rather than a strictly local Cauchy law. Version 0.5 separately removes the numerical ambient-dimension input at the carrier layer, as follows.

3.3 A target-free 1 + 3 carrier and its affine completion

Let the source-derived observer quotient be $Q_{\text{obs}} = V_4$, with intrinsic affine action

$$\text{AGL}(2, 2) \cong S_4.$$

Start over the rational source layer and let $W_{\mathbb{Q}} = \mathbb{Q}^{V_4}$ be the space of observer readouts. Impose three explicit admissibility conditions: the readout contains the constant order unit, separates all four quotient states, and is invariant under the full intrinsic affine action. The S_4 permutation commutant is spanned by I and the all-ones matrix J . Its equivariant idempotents have ranks 0, 1, 3, 4. Unitality excludes ranks 0 and 3, while point separation excludes rank 1. Therefore the minimal admissible readout is the full carrier, without supplying a target dimension:

$$W_{\mathbb{Q}} = \mathbb{Q}\mathbf{1} \oplus W_0, \quad \dim(\mathbb{Q}\mathbf{1}, W_0) = (1, 3). \quad (4)$$

The three-space W_0 is the irreducible mean-zero representation. All 24 affine observer actions preserve this split.

The finite Braid data do not by themselves contain the real numbers as a completed scalar field. We therefore state one completion principle explicitly.

Assumption 3.1 (Observer Uniform Cauchy Closure). Observer readouts and the marked clock direction are closed under uniform Cauchy limits without adding independent observer directions.

Under Assumption 3.1, scalar completion gives

$$W = \mathbb{R}^{V_4} = \mathbb{R}\mathbf{1} \oplus W_{0,\mathbb{R}}, \quad \dim W = 4, \quad (5)$$

with the same 1 + 3 split. Every finite-dimensional real vector space canonically determines an affine smooth carrier $\text{Aff}(W)$: choosing an origin gives a global chart, changing that origin is only a translation, and

$$T_x \text{Aff}(W) \cong W \quad (6)$$

for every x . Thus neither the numeral four nor a separate ambient four-manifold is a primitive mathematical input. A declared cup/null composition additionally gives the dimensionless Lorentz form

$$g_D = \frac{16}{75} \text{diag}(-3, 1, 1, 1), \quad (7)$$

whose four evaluation rays are null. Physical rods, seconds, nonconstant curvature, and the identification of points of $\text{Aff}(W)$ with physical events do not follow from this algebraic completion.

The source path and the carrier have different roles. The path indexes inclusions and disjoint supports in the operator net; it is not renamed as three spatial axes. The three relative directions in Eq. (4) are internal observer-readout directions. The v0.5 physical event identification and soldering obligation was collected under CGR below. Section 3.7 shows that a distinct *formal* event identification is source-native inside the fixed mathematical model, without claiming calibration to physical spacetime.

3.4 The source also closes a finite gravity equation

The source-native construction does not stop at a static geometry. On each certified edge (m, d) it obeys the exact first-order coframe equation

$$E_{m \oplus d} = C_{m,d} E_m R_d + K_{m,d} [C, j, F], \quad (8)$$

with the second covariant face closure

$$D_C K = F_C E. \quad (9)$$

Here $C_{m,d}$ is the source-generated ECDR/BSFG connection, R_d reverses the crossed generator orientation, j is the IBC current, F is the affine face curvature recomputed from (C, j) , and $K[C, j, F]$ is the coframe source that cannot be produced by invertible transport alone.

On the fixed eight-source orbit, Eq. (8) recovers 24/24 future coframes, 24/24 nonzero source terms, and all future ranks exactly; Eq. (9) holds on 12/12 faces and is nonzero on 10/12. Transport without K fails on all 12 rank-changing edges, while zeroing the curvature fails on all 40 nonzero-curvature columns. The source term is therefore not a removable decoration of a stored geometry.

For the induced spatial Gram matrix $G_m = E_m^\top g_m E_m$, the same update yields the finite metric-affine equation

$$\boxed{G_n = R_e^\top G_m R_e + \mathcal{N}_{e,E} + \mathcal{L}_{e,K} + \mathcal{Q}_{e,K}} \quad (10)$$

with

$$\mathcal{N}_{e,E} = R_e^\top E_m^\top (C_e^\top g_n C_e - g_m) E_m R_e, \quad (11)$$

$$\mathcal{L}_{e,K} = (C_e E_m R_e)^\top g_n K_e + K_e^\top g_n (C_e E_m R_e), \quad (12)$$

$$\mathcal{Q}_{e,K} = K_e^\top g_n K_e. \quad (13)$$

All three terms are nonzero and separately necessary on 24/24 edges. Thus the connection's nonmetricity, the source-linear response, and the finite source-quadratic response cannot be collapsed into one another. This is a source-generated finite gravitational dynamics in the declared model. Its packet lifts by one fixed law to every finite braid stage, but only as a stage-compatible cylinder section: it does not yet generate new local spacetime cells.

Equations (8)–(10) are not themselves the Einstein equation $G_{\mu\nu} = \kappa T_{\mu\nu}$. Calling them finite Braid gravity equations records their exact source–geometry dynamics while preserving that distinction. The next construction gives a direct but explicitly conditional local connection to the Einstein equation.

3.5 A conditional source-first local Einstein completion

An initial inverse test showed that the rank-ten UB476 response map can reconstruct a restricted Einstein target to relative error 3.69×10^{-15} while preserving trace and the registered G_1 energy. Because that construction solved backward from the Einstein tensor, it established capacity but not noncircular selection. A stricter comparison then produced an exact no-go: the independently fixed UB476 creation response has

nonzero 01, 02, and 03 components, whereas the restricted ultrastatic target sets all three to zero. Even the best unrestricted scalar fit leaves a relative residual of approximately 0.745.

The successful construction therefore reverses the direction. Let $\Theta_{\mu\nu}$ denote the symmetric ten-component UB476 response fixed before any Einstein target is read. Under the three declared bridges

- (i) $\Theta_{\mu\nu}$ is typed as a local gravitational source tensor,
 - (ii) the independently fixed marked/full source-rank ratio 3/5 is used as the coupling, without fitting it to an Einstein target, and
 - (iii) the source-free Weyl tensor is set to zero as the minimal infrared completion,
- the source first fixes

$$G_{\mu\nu}(p) = \frac{3}{5} \Theta_{\mu\nu} \quad (14)$$

at a base point p . Four-dimensional trace reversal determines the Ricci tensor and scalar curvature; zero Weyl curvature then determines an algebraic Riemann tensor. A standard normal-coordinate construction produces a smooth quadratic Lorentz metric with the required two-jet at p [18].

Every step was checked in exact arithmetic. The Riemann tensor has the two pair antisymmetries, pair-exchange symmetry, and algebraic Bianchi identity; its Ricci and Einstein contractions reproduce Eq. (14); reconstruction from the metric two-jet returns the same curvature. At unit formal amplitude it has 136 nonzero components. The nonzero 0i source components are not discarded: they generate 93 nonzero temporal-derivative entries and 84 nonzero mixed-component entries of the metric two-jet. The base metric is nondegenerate with Lorentz signature, so the quadratic metric retains that signature in a sufficiently small neighborhood. The same source packet and construction apply in all eight signed sectors.

This is a genuine source-first Einstein connection: the source is fixed before the geometry, and the resulting metric satisfies the Einstein relation exactly at p . It is also conditional. The numerical ratio 3/5 is source-derived rather than manually tuned, but interpreting that ratio as a gravitational coupling, typing the response as stress, and choosing a zero-Weyl completion are bridge assumptions. Differential Bianchi propagation, a neighborhood solution of the Einstein PDE, Cauchy dynamics, physical units, and empirical gravity are not obtained. Moreover, the independently certified Braid curvature does not yet select the same complete Riemann tensor: the first coverage audit found only one of the six required bivector slots in a common Levi–Civita carrier, and subsequent audits localized the remaining obstruction to placing all temporal and spatial curvature records in one $\text{End}(T)$ family. This is underdetermination, not a computed disagreement with Eq. (14).

3.6 From pointwise arrival to a conditional continuum field equation

The preceding construction is an *arrival* result: it builds a metric two-jet whose Einstein tensor has the required value at one point. A field-equation *derivation* is stronger. It must define an off-shell action on a neighborhood, vary the admissible fields independently, and obtain the equation for the allowed solutions rather than construct one compatible jet.

The v0.4 formulation listed four bridge laws: temporal soldering, continuum/infrared convergence, a matter/Ward law, and common normalization. Subsequent dependency audits showed that these are not four independent principles. Temporal soldering and continuum convergence are two components of one source-to-geometry realization. Common normalization is irrelevant in vacuum and, with matter, is part of the matter-action provenance rather than a fourth gravitational law. The v0.5 *dependency packet*, retained here as the precise antecedent of its continuum theorem, was therefore:

Assumption 3.2 (Causal geometric realization, CGR). There is a refinement-compatible realization from the completed carrier $\text{Aff}(W)$, the source-path local net, the temporal $\text{Aff}(V_4)$ Cartan link, and the finite Hodge–Palatini cells to local metric-affine fields and regions such that:

- (i) net inclusions and overlaps map to local regions; causal disjointness and tangent/cotangent typing are preserved;
- (ii) the temporal Cartan link and the finite Palatini cells are natural under refinement; and
- (iii) the localized sums admit the uniform compactness and compactly supported first variations required by the continuum Palatini limit.

Assumption 3.3 (Marked matter realization and routing, MMR). The source G_1 Gibbs/modular response and the doubled density–polarization functional realize one local covariant metric-only matter action on the CGR geometry. The same source trace routes the full sector to gravity and the source rank-three marked sector to matter before target comparison. Consequently,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}, \quad \kappa_B = \frac{3}{5}, \quad (15)$$

the on-shell Ward identity holds, and matter hypermomentum vanishes in this selected metric-only effective class.

Under CGR and MMR the working infrared action is

$$S_{\text{IR}}[g, \Gamma, \psi] = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \psi]. \quad (16)$$

Metric or coframe variation gives

$$R_{(\mu\nu)}(\Gamma) - \frac{1}{2} g_{\mu\nu} R(\Gamma) = \kappa_B T_{\mu\nu}. \quad (17)$$

The connection equation gives $\Gamma = \Gamma_{\text{LC}}(g)$ in the Lorentz branch and Levi–Civita modulo a projective one-form in the metric-affine branch. The projective shift changes only the antisymmetric Ricci tensor, so it cancels from $R_{(\mu\nu)}$, the Palatini scalar, and Eq. (17). Substitution, followed by the MMR routing, yields

$$\boxed{G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}} \quad (18)$$

as an Euler–Lagrange equation on a continuous spacetime neighborhood.

Theorem 3.4 (v0.5 conditional continuum Einstein theorem). *The exact finite source results of this paper, together with CGR and MMR, imply the continuum-neighborhood field equation (18) in the long-wavelength and low-curvature sector of the selected metric-only matter class. This is a conditional variational derivation and is strictly stronger than the pointwise arrival result (14).*

At the v0.5 cutoff neither CGR nor MMR was derived in full from the pointed-braid source. The finite side nevertheless reached farther than v0.4 recorded: a source-derived rank-four metric-response carrier persists through the inductive tower; the scalar response is integrated by the canonical Reynolds-projection GKSL semigroup; the three relative directions are inner-derivation controls; and their sum is a single CPTP hybrid evolution. On the source-path interval net this evolution is strictly local on its marked support and compatible with right-tail refinement. These results supply algebraic locality and evolution. The next subsection reports the *post-v0.5* source-event and matter-action developments without retroactively attributing them to this theorem’s original proof.

The four response directions are locally independent but not four commuting coordinate flows. Their first Lie-bracket closure exits the rank-four span. This prevents a first-class interpretation of those particular flows, but it does not obstruct the single hybrid GKSL generator or the conditional Palatini variation. At this historical cutoff the finite G_1 Gibbs state supplied an exact algebraic KMS seed and the rank ratio $3/5$ was structural, while a local doubled α -metric action and marked gravity–matter routing were still requirements.

The two-principle count was minimal relative to the v0.5 audited source inventory. CGR did not supply a matter action or its relative normalization; MMR did not supply event localization, curvature, or continuum Palatini variation. The formal status of the first principle, and the conditional status of the second, have since changed as follows.

3.7 v0.6.1 code audit: finite events versus the continuum arrow

The source-selected one-site marker M_{ev} has simple spectrum $(-7, -5, -4, -1, 7)$ in all eight signed sectors. Its rank-one spectral projectors E_c generate event atoms $C_{cd} = E_c \otimes E_d$, rather than inheriting labels from an ambient spacetime. The actual signed braid crossing permutes all 25 atoms exactly: nine are fixed and sixteen form eight transpositions. The induced table differs from both identity and ordinary tensor flip, and its three-site action satisfies the set-theoretic Yang–Baxter relation. A cylinder event equals the sum of its five children, and the co-moving observer reads the same event back after transport. This is the exact fixed-source content of BGCE139.

The code-level audit separates three arrows which the earlier source reports grouped under “formal CGR retirement.” BGCE139 checks the actual finite event action. BGCE138 constructs a smooth C^2 field neighborhood around a flat anchor and describes edge integration and holonomy sampling from any field in that chosen neighborhood. It does *not* read the actual finite Braid coframe/connection sequence and bound its distance to one smooth limiting field. BGCE099 states the Palatini convergence and first-variation implication under UTC; its evaluator records the standard equations and a projective antisymmetry check rather than verifying an actual source sequence. Thus the finite event component of CGR is supported, while its source-to-continuum component is still a separate hypothesis. The earlier internal “PASS” label cannot substitute for the missing reverse arrow.

CGR obligation	Current direct code witness	Status for this paper
Spectral finite events	Actual eight-sector matrix conjugation, 25 transitions, and 125 Yang–Baxter triples (BGCE139).	Checked, fixed finite model
Source-local address	Marker spectrum, projectors, 625-child combinatorics; chosen four-base-five quotient (BGCE138).	Mathematical construction
Smooth off-shell fields	Explicit flat anchor, diagonal-dominance bound, and compact-support bump; smooth-to-edge sampling is described (BGCE138).	Existence, not source dynamics
Actual source $\rightarrow C^2$ limit	No stagewise comparison of actual E_m, U_m with a common limiting field, uniform difference bound, branch margin, or first-variation error.	Open
Physical event and scale	No natural rod, clock, curvature, or coupling calibration.	Open

In particular, a map from smooth fields to consistent discrete samples does not establish the converse map from the actual finite Braid data to a uniformly regular smooth field. This is a proof gap, not a counterexample to the possible existence of such a limit. The conditional continuum theorem above remains valid under its stated CGR hypothesis; CGR has not yet been removed as an independent premise for that theorem.

3.8 v0.6: local full matter action, readout, and conditional MMR2

BGCE142 fixes ten original symmetric matter-source operators F_A (where A ranges over the ten pairs $0 \leq \mu \leq \nu \leq 3$) before any gravitational target is compared. BGCE143 uses their full-trace analytic generating functional $W(a)$ to define a normalized doubled matter action on a selected smooth source-cylinder field domain. The code checks normalized-tail identities and child-constant 625-fold coarsening;

the continuum integral and stress variation follow analytically when continuous local sources and a metric field are supplied. This constructs formal *MMR1* on that domain, not actual Braid-driven nonconstant field convergence, physical coframe/Ward identification, or a source-selected material boundary link. Doubled influence functionals and quantum Markov conditions have substantial prior art [3, 8]; the claimed source-specific content here is the fixed G_1 operator family and its declared cylinder typing, not those generic techniques.

The event readout is not limited to the first derivative at $a = 0$. In the fixed finite model, the ten F_A lie in the operator span of the same two-direction rotated marker-event probabilities through mixed degree three plus the four G_1 energy probabilities (BGCE185). Since this is an operator identity, it holds at every tilted Gibbs state ρ_{ta} . With $W(0) = 0$ one obtains the exact reconstruction

$$W(a) = \int_0^1 \sum_A a_A \text{Tr}(\rho_{ta} F_A) dt. \quad (19)$$

Thus the *unchanged* full nonlinear $W(a)$ has a mathematical readout representation if the entire tilted Gibbs family ρ_{ta} is available (BGCE187). BGCE185 checks finite operator-span membership, but explicit readout coefficients for all ten operators have not been output; BGCE187 is an analytic integration argument, not an independently executable all- a verifier. A single-state readout, physical instrument, and marked-action choice do not follow.

The remaining effective-action routing has a constructive *conditional* answer. Let $P = P_2$ be the local rank-75 head mark, $p = P_{\text{src}}$ the outer rank-three mark, $Q = I - P$, and $q = I - p$. State the extra relational mark link explicitly as

$$L_{\text{RML}} = P \otimes p + Q \otimes q, \quad X_a = W(a)P. \quad (20)$$

For every a , including nonlinear source directions,

$$L_{\text{RML}}(X_a \otimes I_5)L_{\text{RML}} = W(a)P \otimes p, \quad (\tau_P \otimes \tau_5)(W(a)P \otimes p) = \frac{3}{5}W(a), \quad (21)$$

where $\tau_P = \text{Tr}_{125}/75$ on the P corner and $\tau_5 = \text{Tr}_5/5$. The value $3/5 = \text{rank}(p)/5$ is not fitted to the Einstein equation, but it is *definitionally forced once the RML head unit, same-direction link, and normalized traces have been selected*. Within that stipulated support law, the outer-bare $W(a)P \otimes I_5$ is excluded whenever $W(a) \neq 0$. The distinct head-bare competitor $W(a)I_{125}$ is not excluded by the source alone; choosing $X_a = W(a)P$ is part of RML. BGCE197 computes an actual finite current witness for the projection identity but carries arbitrary $W(a)$ as a scalar coefficient. Equation (21) is a valid *conditional effective-action identity for MMR2*, not independent evidence that RML is selected or that $3/5$ is a physical gravitational coupling.

The condition matters. Neither the original F_A nor the whole Gibbs trace lives in the P corner; compressing before the trace changes the original W (BGCE200/205). The same covariance and refinement laws also admit the reverse link $P \otimes q + Q \otimes p$, whose outer cup is $2/5$, unless an additional source relation selects orientation. Moreover, applying the one-shot head effect pointwise on the existing four-address cylinder fails to give $3/5$ for arbitrary cellwise W (BGCE199). Finally, the sharp marker-event readout and P are noncommuting: an event-first nonselective readout leaves weight $1/2$ on an unmarked digit, while mark-first inserts P by construction (BGCE206). These are scoped counterexamples to overreading the conditional result, not retractions of Eq. (21).

The revised dependency statement is therefore precise. The finite event-algebra part of CGR is checked; full formal CGR retirement is not established by the present code because the actual source-to-smooth limit is missing. Formal MMR1 is constructed on a selected smooth cylinder; MMR2's full- W effective routing is an identity conditional on RML. The source-only selection of RML, its physical local implementation, a source-selected coframe and on-shell matter Ward identity, and common gravity-matter action provenance remain open. The earlier continuum theorem still follows under its stated realization assumptions; no extra factor of $3/5$ may be inserted a second time.

4 Post-v0.6.1 assumption removal and Lorentz-carrier derivation

This section records results obtained after the v0.6.1 code audit. It does not erase the historical statements above. In particular, the RML calculation remains a valid conditional identity and the source-to-smooth convergence gap remains open. The new result is that the fixed source packet supplies a stronger selection principle than RML alone, and that the same marked event lineage supplies a four-direction Lorentz carrier. The logical update is therefore selective:

1. MMR2 closes inside the declared fixed event-conditioned model and complete source-atom partition class;
2. the relative factor 3/5 is inherited from the selected normalized source cup rather than manually chosen;
3. the four-direction event carrier and its Lorentz signature are source-derived in a stated naturality class;
4. the actual source-to-smooth convergence theorem remains open; and
5. the parent action, physical stress pullback, and Ward identity remain compressed in SDPC rather than counted as derived.

4.1 Fixed-packet selection closes MMR2 in the declared model

Let the event-conditioned corner packet be

$$\mathcal{S}_\cap = \{PG_1P, PF_AP \mid A = 1, \dots, 10\}. \quad (22)$$

The three marked source atoms generate exactly five partition retractions: the full partition, three two-block partitions, and the fully separated partition. Earlier rank tests showed that all five retain a ten-dimensional source span and a nondegenerate BKM Hessian. That fact did not establish physical equivalence, because a nontrivial retraction replaces the fixed Hamiltonian and response operators by dephased operators.

The sharper no-retuning test asks which retraction preserves every operator in Eq. (22) exactly. In all eight signed sectors, only the full partition does so. The first changed operator for every nontrivial partition is $H_\cap = PG_1P$; the corresponding coherence-residual ranks are 28, 32, 32, and 46 for the four non-full partitions. No coefficient fit, matrix exponential, or result-dependent adjustment enters this comparison.

Proposition 4.1 (Fixed-model MMR2 closure). *Inside the declared event-conditioned source model and the complete five-element partition-retraction class generated by the three marked source atoms, exact preservation of the fixed eleven-operator packet selects the full corner uniquely. The full-corner matter action and its routing therefore do not require an additional nondemolition selection law within this class.*

The scope is important. Proposition 4.1 does not prove uniqueness among arbitrary completely positive instruments, arbitrary source subalgebras, or possible natural measurement devices. A dephasing instrument can exist as a different state-update rule without authorizing replacement of the already fixed Hamiltonian by a dephased Hamiltonian. The theorem selects the matter packet needed by the fixed model; it does not classify all quantum instruments.

The normalized tail trace sends the corner unit to the outer event effect, and the selected source-cup composition gives

$$\kappa_B = \frac{3}{5}. \quad (23)$$

This updates the v0.6.1 dependency language. The value in Eq. (23) is no longer conditional on choosing RML as an otherwise unselected competitor inside the audited fixed-packet class. It remains a relative source normalization, not an empirically calibrated Newton constant, and it does not by itself establish a physical stress tensor.

4.2 Actual Braid event transition generates a rank-four base dynamics

The actual event update is not the identity or the ordinary tensor flip. For each adjacent pair of source directions, its exact displacement second moment is

$$M_{\text{pair}} = \begin{pmatrix} 28 & -20 \\ -20 & 28 \end{pmatrix}. \quad (24)$$

It has eigenvalues 8 and 48 and rank two. The identity comparator has rank zero and the ordinary flip comparator rank one. Thus the full pair coupling is specific to the actual non-flip event transition within this comparator class.

Average Eq. (24) over the 24 source charts. With

$$P_0 = \frac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad P_1 = I_4 - P_0, \quad (25)$$

the result is

$$Q_4 = \frac{104I_4 - 20\mathbf{1}\mathbf{1}^\top}{50} = \frac{12}{25}P_0 + \frac{52}{25}P_1. \quad (26)$$

Q_4 has diagonal entries $42/25$, off-diagonal entries $-2/5$, and eigenvalues $12/25$ on the uniform line and $52/25$ on the standard three-dimensional complement. It is positive and rank four. Neither an external axis weight nor a preassigned physical $1 + 3$ ordering is used.

The source response is the derivative of its log-partition function. Within the response-exact Bregman class, exactness of the response-mismatch one-form and vanishing on the diagonal select the log-Z Bregman divergence, equivalently the Umegaki relative entropy of the Gibbs family. The neighboring-edge quadratic term is

$$\delta^2 S_{\text{edge}} = \frac{1}{2} \int Q_4^{ab} \langle \partial_a \lambda, \partial_b \lambda \rangle_{\text{BKM}} d\mu. \quad (27)$$

The base moment has rank four and the internal tested BKM metric rank four, giving a rank-sixteen gradient quadratic form. This closes source-independent local-label dynamics on the fixed cylinder. Equation (27) is positive; it is not yet the Lorentzian physical action.

4.3 Four source-null links reconstruct the Lorentz metric

Let L be the matrix whose columns are the four certified future-null source links. The exact result is

$$\det L = \frac{81}{16} \neq 0. \quad (28)$$

The source-null Gram matrix G_{null} has zero diagonal and off-diagonal entries $-12/25$. Therefore

$$g = L^{-\top} G_{\text{null}} L^{-1} = \text{diag}\left(-\frac{16}{25}, \frac{16}{75}, \frac{16}{75}, \frac{16}{75}\right) \quad (29)$$

is reconstructed uniquely from the four links. Its signature is $(3+, 1-)$. The induced symmetric-square map $\text{Sym}^2(L)$ has rank ten, so the four-direction carrier is algebraically nondegenerate on all ten symmetric rank-two components.

This result strengthens the four-dimensional claim without making it universal. Four operational directions are generated and independent in the fixed source model. The paper still does not prove that every admissible Braid source must generate four dimensions, nor that the actual finite field sequence converges to a global four-manifold.

The positive event moment remains positive under the null coframe:

$$LQ_4L^\top = \text{diag}\left(\frac{27}{25}, \frac{117}{25}, \frac{117}{25}, \frac{117}{25}\right). \quad (30)$$

Sylvester's law of inertia therefore excludes a real invertible coframe congruence as the origin of the Lorentz sign. A null tetrad can reconstruct the Lorentz metric while still failing to convert a positive Dirichlet action into a Lorentzian kinetic action. These are different statements.

4.4 Source causal grading yields the Lorentz event carrier

The source causal analysis independently selects the uniform sum of the four future-null rays as the timelike mode. Its reflection on the ray carrier is

$$J_{\text{ray}} = I_4 - \frac{1}{2} \mathbf{1} \mathbf{1}^T = -P_0 + P_1. \quad (31)$$

It has eigenvalue -1 on the uniform line and $+1$ on the standard complement. Since Eqs. (26) and (31) are simultaneously S_4 -equivariant, they commute. Their product is

$$K_{\text{evt}} = J_{\text{ray}} Q_4 = -\frac{12}{25} P_0 + \frac{52}{25} P_1. \quad (32)$$

It is symmetric with exact signature $(3+, 1-)$. Its diagonal entries are $36/25$ and its off-diagonal entries $-16/25$.

Theorem 4.2 (Source-derived Lorentz event carrier). *In the declared fixed source model, the actual Braid event moment and the independently source-selected causal grading determine the Lorentz-signature carrier $K_{\text{evt}} = J_{\text{ray}} Q_4$. No Wick rotation, manual sign assignment, or fitted continuous coefficient is used.*

The tensor type fixes the correct normalization target. Since $Q_4^{ab} \partial_a \lambda \partial_b \lambda$ is contravariant, it must be compared with G_{null}^{-1} rather than G_{null} . Restrict a normalizer to the positive, self-adjoint, S_4 -equivariant class

$$A_{\text{kin}} = a_0 P_0 + a_1 P_1. \quad (33)$$

The equation

$$A_{\text{kin}}^T K_{\text{evt}} A_{\text{kin}} = G_{\text{null}}^{-1} \quad (34)$$

has the unique positive solution

$$a_0 = \frac{25\sqrt{3}}{36}, \quad a_1 = \frac{25\sqrt{39}}{156}. \quad (35)$$

All 24 chart permutations commute with the projector decomposition, and the character frame recovers

$$g^{-1} = \text{diag}\left(-\frac{25}{16}, \frac{75}{16}, \frac{75}{16}, \frac{75}{16}\right). \quad (36)$$

The uniqueness statement is restricted to the declared positive self-adjoint S_4 -equivariant class. It is not a claim of uniqueness under arbitrary Lorentz isometries or nonlinear field redefinitions.

4.5 Action-selection no-go results

Theorem 4.2 does not establish that the retained discrete variational principle selects Eq. (32). The certified Umegaki edge action is a positive uniform sum and contains no J_{ray} insertion. Two plausible shortcuts were tested exactly.

First, forward and reverse event displacements have identical second moments. Even and odd chart averages also equal Q_4 . Their orientation-odd difference is zero, and every scalar combination of forward and reverse moments is a scalar multiple of Q_4 . The Braid word retains orientation, but the quadratic relative-entropy Hessian erases it.

Second, the closed-time-path branch form

$$\eta_{\text{br}} \otimes Q_4, \quad \eta_{\text{br}} = \text{diag}(1, -1), \quad (37)$$

has signature $(4+, 4-)$ on branch $\otimes$ base space. Every contraction with a fixed branch vector v gives

$$(v^T \eta_{\text{br}} v) Q_4, \quad (38)$$

which is only a scalar multiple of the positive base moment. It cannot produce the mode-selective $(3+, 1-)$ form. A branch-to-ray intertwiner chosen to do so would merely rename the missing assumption.

These failures preserve rather than weaken the carrier theorem. They show that carrier derivation and action derivation are distinct gates, and they prevent the Lorentz signature from being counted twice.

4.6 Source-Dynamical Parent Completion

After v0.6.1 the remaining physical-stress solder and Ward law were compressed into one compound parent-action principle.

Assumption 4.3 (Source-Dynamical Parent Completion, SDPC). On the same source cylinder there exists one local parent matter action $S_{\text{parent}}[g, \Psi; a]$ satisfying all of the following:

1. its source first jet reproduces the derived full-corner response and Proposition 4.1;
2. its metric/coframe coupling fixes the pullback of the ten source derivatives, without an externally selected solder or free response scale;
3. its physical section $a = 0$ contains nontrivial source-independent matter dynamics;
4. its kinetic principal symbol selects Eq. (32) at action level rather than inserting J_{ray} after variation;
5. it is invariant under the diagonal diffeomorphism/frame action, so its off-shell Noether identity relates stress divergence to the matter Euler–Lagrange terms;
6. the same matter fields are placed on shell by this action, yielding covariant stress conservation; and
7. its connection sector is compatible with the metric-only or zero-hypermomentum Palatini sector used in Theorem 3.4.

SDPC is one compound existence statement, not a collection of tunable numerical parameters. Under SDPC, the physical stress is defined by the same parent action,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{parent}}}{\delta g^{\mu\nu}}, \quad (39)$$

and the Ward identity follows from its covariance rather than being postulated independently. Together with the retained source-to-smooth convergence and first-variation hypothesis, Eq. (23) then gives Eq. (18).

Several ingredients of SDPC are now source-derived: the fixed source first jet, full-corner routing, local-label dynamics, response-exact edge functional, causal grading, Lorentz carrier, and its natural normalization. SDPC as a whole is not derived. The missing conjunction is action-level selection of the signed grading, the unique physical metric/coframe pullback of the actual operator packet, nontrivial physical-section stationarity, and the associated Ward closure. Thus the unconditional Braid-only Einstein equation remains open.

5 From finite updates to source-generated locality

5.1 A finite source-graph local net

At stage n , let G_n be the graph whose vertices are tensor factors and whose edges join adjacent noncommuting Braid generators. In every certified signed sector, distant generators commute exactly and adjacent

triples obey the Yang–Baxter relation. For a vertex set U , define

$$\mathfrak{A}_n(U) = \text{End}((\mathbb{C}^5)^{\otimes U}) \otimes I_{U^c}. \quad (40)$$

Then $U \subseteq V$ implies $\mathfrak{A}_n(U) \subseteq \mathfrak{A}_n(V)$, disjoint supports commute, and a depth- t adjacent-Braid update expands support by at most graph distance t . The refinement j_n is strict and unital. These are exact finite isotony, locality, and finite-propagation statements on a source-generated graph. They are not yet Haag–Kastler locality on a Lorentzian manifold, and graph distance is not identified with physical distance.

The positive Braid generator candidate

$$L_n^B = \sum_i (I - \text{Ad } R_i) \quad (41)$$

has a refinement-boundary defect

$$\Delta_n(X) = L_{n+1}^B j_n(X) - j_n L_n^B(X). \quad (42)$$

Its normalized boundary reduction is nonzero in all eight sectors, has 41 nonzero matrix entries and squared Frobenius norm 108/5, fills all six registered boundary-response channels, and returns to zero after the inverse crossing. Thus the stable carrier and the local generation event are distinct outputs of the same source lineage.

5.2 A finite Ward law before physical stress-energy

The boundary response, the existing IBC edge data, and the ordered-edge memory admit a six-component cell variable. If d_n is the incidence matrix of the stage path, $D_n = d_n \otimes I_6$, and H is the exact rank-six cell Gram matrix, define

$$\mathcal{S}_n(q) = \frac{1}{2} (D_n q)^\top (I_{n-1} \otimes H) (D_n q), \quad L_n = D_n^\top (I_{n-1} \otimes H) D_n. \quad (43)$$

For every $n \geq 2$,

$$\text{rank } L_n = 6(n-1), \quad \ker L_n = \mathbf{1}_n \otimes \mathbb{F}^6, \quad (\mathbf{1}_n^\top \otimes I_6) L_n = 0. \quad (44)$$

The last identity is an exact all-finite-stage constant-shift Ward law. For every graph interval, interior edge fluxes cancel and only the two cut-boundary fluxes remain. The incidence mechanism is not uniquely Braid-specific; the Braid-specific content lies in the rank-six cell packet and its actual-versus-flip response. Equation (44) is weaker than stress-energy conservation and does not constitute a spacetime Bianchi identity.

6 Continuous cells and the native metric-affine sector

6.1 Stable-square gluing, metric faces, and curvature

The finite edge transports can be glued on stable squares with exact agreement on overlaps. On the source-native cubical cells, the generated Gram data admit continuous face representatives, and their compatible edge transports define a continuous connection with nonzero rank-four curvature on the certified face packet. A common projective timelike line and a $1 \oplus 3$ splitting can then be selected consistently on the regular sector. The induced spatial connection retains nonzero curvature.

These statements produce continuous representatives on certified finite cells; they do not derive a global smooth manifold, an arbitrary-field connection, or a physical continuum limit. A separate refinement analysis also shows why this distinction matters: the current tensor-tower embedding preserves packets but does not subdivide geometric edges, while the exact affine edge refinement is endpoint-blind to the nonuniform off-shell defects needed to select an action by refinement alone.

6.2 Torsion and nonmetricity are not removable errors

The full source connection splits into a metric-compatible polar core and a metric-symmetric part. Both curvature components scale at the same leading area order on the presently covered regular faces. Neither the tested perturbative contortion correction nor the exact finite single-face torsion-free ansatz closes. The unmodified construction therefore does not converge to the Levi–Civita connection.

This negative result selects a wider native geometry rather than eliminating gravity. Write the regular-face source coordinate as

$$\hat{K} = I - A, \quad A = UP. \quad (45)$$

With $\epsilon = \text{sign det } U$, $R = \epsilon U$, $\rho = (\det P)^{1/3}$, and $S = \rho^{-1}P$, one has the unique reconstruction

$$A = \epsilon R \rho S, \text{quad } \det S = 1. \quad (46)$$

The connected sectors have dimensions

$$\underbrace{3}_{\text{rotation}} + \underbrace{1}_{\text{dilation}} + \underbrace{5}_{\text{unimodular shear}}, \quad (47)$$

together with the discrete parity ϵ . All three continuous sectors are nontrivial in all 16 regular-face records, and their product reconstructs A exactly. This is precisely the representation type expected for a post-Riemannian affine source. The names “spin,” “dilation current,” and “shear hypermomentum” are not physical identifications here; no matter current has yet been measured or derived.

6.3 Stratified junctions

Rank-changing paths do not destroy the entire tangential metric. At the four exceptional junction records, after removal of the declared endpoint double zero, the determinant is exactly a nonzero constant times $(t - r)^2$ over $\mathbb{Q}(\sqrt{17})$, with $0 < r < 1$ in both real embeddings. At r the full Gram matrix has rank two and inertia $(0, 2, 1)$: two positive directions and one zero direction, with no negative direction. The metric is positive definite on both adjacent open intervals. The isolated event is therefore a codimension-one channel loss with a positive two-dimensional quotient, not a collapse of the tangential sector. Interpreting it as physical dimension creation would require an additional bridge and is not claimed.

7 A source-native finite first-order action

7.1 Why a metric-only action is insufficient

The full Braid boundary load decomposes into zero clock response and nonzero rotation, dilation, and shear responses. A functional depending on the spatial coframe E only through the Gram matrix $E^T g E$ has variation in the six-dimensional image

$$\{gES : S^T = S\}. \quad (48)$$

The rotation response lies instead in the independent three-dimensional image

$$\{gER : R^T = -R\}. \quad (49)$$

The two images intersect only at zero, while the certified full load has nonzero components in both. Therefore the source response cannot be generated by a metric-only action. Coframe and connection variables must remain independent at the finite level.

7.2 The full-trace action

For each oriented face f , let $\mathcal{R}_f$ denote the history-dressed linear curvature obtained from the two ordered boundary paths, let E_f denote the corresponding source-generated base coframe, let $E_\top$ be the common-top coframe, and let $g_\top$ be the generated top metric. The source-native full-trace action is

$$\mathcal{S}_B[E, C, g] = \sum_{f \in \partial e} \sigma_f \text{Tr}(E_\top^\top g_\top \mathcal{R}_f[C] E_f), \quad (50)$$

where σ_f is the orientation fixed by the cell boundary. Equal column weights are forced, up to one overall scale, by covariance under the generating permutations of the three spatial columns. No face weight, sector weight, prefix-dependent scale, X51 residual, or Einstein tensor is used.

On the certified packets, the action is nonzero for both prefixes. Its analytic coframe variation agrees with exact central differences in 192/192 components, and its analytic connection variation agrees in 384/384 components. The top coframe Euler response reproduces the independently frozen full boundary solder for both prefixes, including the rotation, dilation, and shear sectors. The connection Euler response is nonzero on all 12 active edges for each prefix. Thus one finite action produces both coframe and connection equations.

The action is not a fitted scalarization of the affine translation block. Affine-origin invariance eliminates the translation scalar and selects the linear coframe-curvature pairing up to normalization. Earlier direct attempts to identify a face scalar, a Cartan three-cell scalar, or a full curvature solder with the prior X51 residual all failed by exact non-collinearity. Those no-go results are retained: Eq. (50) is justified by its own source covariance and variation, not by reverse-engineering X51.

7.3 Exact local-frame Noether identity

Let $\mathcal{E}_E$, $\mathcal{E}_C$, and $\mathcal{E}_g$ denote the coframe, connection, and top-metric Euler responses of Eq. (50). Under a local infinitesimal frame change at a vertex, the incident coframes and connections transform together. Invariance of the action gives the matrix identity

$$\mathcal{N}_v(\mathcal{E}_E, \mathcal{E}_C, \mathcal{E}_g) = 0 \quad (51)$$

at every vertex, where $\mathcal{N}_v$ is the signed sum of the coframe contribution, incident incoming and outgoing connection contributions, and, at the top vertex, the metric contribution. The identity holds exactly at all 14 non-top vertex tests and both top-vertex tests. With $g_\top$ fixed, the full $GL(4)$ identity fails as it should, while all 12 tested generators preserving $g_\top$ retain the Lorentz-type subidentity. Finite local-frame covariance also passes 16/16 tests.

Equation (51) removes the previous absence of an action-level conservation structure. It does not make the Euler responses vanish. A Noether identity relates field equations; it is not itself a solved field equation or a physical energy current.

7.4 A covariant-defect variational connector

The existing IBC, ECDR, BODC, and history-curvature library was tested as a possible independent balance action. A fixed opposite source cancels the action value, top-coframe response, and metric response at the evaluated point, but leaves eight base-coframe masks and 24 connection-edge responses. Recomputing the opposite term dynamically cancels everything only because it makes the total action identically zero. That is a tautological negative copy, not an independent physical source.

The resulting no-go did not establish that the Braid source lacks a variational connector. On an oriented edge $e = (m, d)$ with high vertex $n = m \oplus d$, define the independently generated covariant coframe defect

$$K_e = E_n - C_e E_m R_d. \quad (52)$$

This is the same X36 defect reconstructed from the source on all 24 certified edges. Its coefficient-one quadratic action is

$$\mathcal{S}_D[E, C, g] = \frac{1}{2} \sum_e \text{Tr}(K_e^\top g_n K_e). \quad (53)$$

It was frozen without reading the curvature-action Euler residual, X51, or any Einstein tensor.

The connector is nontrivial and off shell. Its exact analytic derivatives agree with central differences in 832/832 tested entries across coframe, connection, and metric slots. The induced map has rank 12 on every full defect space and rank 8 on every traversed-column-zero source subspace; the actual cotangent image is nonzero on 24/24 edges. Finite local-frame covariance holds in 16/16 tests, the infinitesimal vertex Noether identities hold in 256/256 tests, and both off-shell perturbations change the action. Thus Eq. (53) supplies the previously missing source-to-cotangent typing without fitting a negative copy.

The current result is therefore

$$\boxed{\text{Braid source} \rightarrow \text{geometry} \rightarrow (\mathcal{S}_B, \mathcal{S}_D) \rightarrow T_{(E,C,g)}^*, \quad \delta(\mathcal{S}_B + \mathcal{S}_D) = 0 \text{ untested.}} \quad (54)$$

The finite-action frontier has therefore moved. The connector no longer remains open; what remains open is whether the independently frozen coefficient-one sum $\mathcal{S}_B + \mathcal{S}_D$ makes all coframe, connection, and metric Euler responses vanish on the actual packet. Passing that test would give a finite source-balanced field equation without coefficient fitting. Failure would classify the residual sector needed for any further coupling. Neither outcome has yet been computed. The separate continuum theorem, Eq. (18), remains conditional on actual source-to-smooth convergence, first-variation control, and SDPC. The post-v0.6.1 fixed-packet theorem closes MMR2 only in its declared event-conditioned class; neither that result nor the Lorentz-carrier identity converts the continuum theorem into a limit of solved finite stationary systems.

8 Why scalar invariants are not enough

8.1 Soldered source holonomy

For each prefix $p \in \{A, B\}$, let

$$E_p \in \text{Mat}_{4 \times 3}(\mathbb{R}), \quad \text{rank } E_p = 3,$$

be the source-generated coframe, let $g_p \in \text{Mat}_{4 \times 4}(\mathbb{R})$ be the ambient symmetric form, and let $H_p \in GL_3$ be the internal face holonomy. Define the Gram matrix

$$M_p = E_p^\top g_p E_p, \quad (55)$$

assumed invertible on the certified regular stratum.

The source-only update has two independently meaningful descriptions. The internal path mismatch is H_p , while the ambient coframe curvature is the difference between the two source-generated paths to the same face. Exact certificates show, in every one of the eight face packets, that both paths reach the same target coframe while the internal holonomy is nonidentity, and that the internal action, ambient curvature action, and source curl agree. A zero-source control fails in all eight packets.

This is not yet Riemann curvature or Einstein gravity. It is a finite source-only holonomy/curvature identity on a regular stratum. Its role here is to provide directional data that is both nontrivial and traceable to the braid source.

8.2 The scalar route and its exact obstruction

Conjugacy-invariant characteristic data of H_A and H_B provide three scalar channels per prefix. The product of their characteristic polynomials gives six nonleading coefficients that are invariant under exchange $A \leftrightarrow B$ and distinguish both one-sided collapses in the certified packets. This is a valid relational readout, but it discards direction.

The loss is not merely aesthetic. For the declared signed-permutation frame group acting on Sym_3 , the fixed subspace is one-dimensional, spanned by the identity. Therefore any full-frame-equivariant linear map from six trivially transforming scalar channels into one anisotropic Sym_3 block has rank at most one. The exact intertwiner calculation has constraint rank 30, a six-dimensional map space, and maximum anchor rank one. Moreover, the original modular residues have no unique real lift and carry no stage index.

Proposition 8.1 (Scalar-anchor no-go). *Within the declared linear, signed-permutation-equivariant anchor class, six presentation-invariant scalar channels cannot determine a full-rank anisotropic symmetric three-tensor. A covariant directional carrier must be formed before the directional information is quotiented away.*

The proposition is scoped. It does not forbid nonlinear maps, separately typed scalar observables, or enriched covariant carriers. It instead identifies the exact repair used next.

9 The generated tensor carrier on the source-derived 1 + 3 type

9.1 Construction

The formulas below were originally written in a declared $\mathbb{R}^4$ basis. In v0.5 their coordinate-free type is the source-derived carrier W of Eq. (5): $E_p \in \text{Hom}(\mathbb{R}^3, W)$, $g_p \in \text{Sym}^2(W^*)$, and $T_C \in \text{Sym}^2(W)$. A chosen constant-plus-character basis identifies these spaces with the displayed 4×3 and 4×4 matrices. Thus the component calculations are unchanged while the numeral four is no longer a primitive target input.

Retain the holonomy defect until it has been soldered to the ambient coframe:

$$D_p = E_p(I - H_p). \quad (56)$$

Define

$$T_p = D_p M_p^{-1} E_p^\top + E_p M_p^{-1} D_p^\top \in \text{Sym}_4(\mathbb{R}), \quad (57)$$

and define the joint or intersection carrier

$$T_C = T_A + T_B. \quad (58)$$

The construction performs the quotient in the correct order. The inverse Gram matrix removes internal-basis dependence, while the coframe preserves ambient direction. Symmetrization yields a rank-two ambient carrier. Only then are the two standpoint contributions joined.

9.2 Transformation laws

Let $G_p \in GL_3$ be independent internal changes of basis. Under

$$E_p \mapsto E_p G_p, \quad H_p \mapsto G_p^{-1} H_p G_p, \quad (59)$$

we have $D_p \mapsto D_p G_p$ and $M_p \mapsto G_p^\top M_p G_p$. Substitution in Eq. (57) gives

$$T_p \mapsto T_p. \quad (60)$$

Thus the carrier does not depend on either prefix's internal presentation.

Let $L \in GL_4$ be an ambient basis change, with

$$E_p \mapsto LE_p, \quad g_p \mapsto L^{-T} g_p L^{-1}. \quad (61)$$

Then M_p is unchanged and

$$T_C \mapsto LT_C L^T. \quad (62)$$

This is covariance, not componentwise invariance: the object is stable while its components change tensorially.

Finally, exchanging A and B leaves Eq. (58) unchanged. Crucially, exchange invariance is not identification of the two inputs. The certified prefix carriers are distinct, and replacing the pair by (A, A) , (B, B) , or identity holonomy changes the result.

Proposition 9.1 (Structural carrier properties). *Whenever M_A and M_B are invertible, T_C is symmetric, invariant under independent internal GL_3 presentation changes, invariant under prefix exchange, and covariant by ambient congruence.*

Proof. Symmetry follows directly from Eq. (57). The internal transformation rules imply

$$D_p G_p (G_p^{-1} M_p^{-1} G_p^{-T}) G_p^T E_p^T = D_p M_p^{-1} E_p^T,$$

and similarly for the transpose term. Exchange invariance follows from commutativity of addition. Ambient covariance follows by substituting LE_p and LD_p into Eq. (57). $\square$

9.3 Exact finite certificate

Theorem 9.2 (Certified joint-carrier theorem). *For the registered actual source packets and their source-generated geometry:*

1. T_A and T_B are nonzero symmetric rank-three carriers;
2. T_C is nonzero, symmetric, and rank three;
3. all ten native components of $T_C \in \text{Sym}_4$ are nonzero;
4. T_C is exactly invariant under independently selected internal GL_3 gauges and under A/B exchange;
5. the congruence law in Eq. (62) holds exactly;
6. T_C differs from $2T_A$, $2T_B$, and the identity-holonomy zero control;
7. all irrational $\sqrt{17}$ parts cancel, so T_A, T_B, T_C descend exactly to rational matrices.

The theorem is an exact fixed-finite result, not a fitted numerical approximation. Items 4–6 jointly matter for the interpretation. Presentation redundancy disappears, ambient direction remains, and neither source standpoint is dispensable.

The ten components have the same mathematical type as an existing ten-component metric-response interface in the larger unification program. Equality to that response, physical normalization, units, insertion into the 87-component state, and laboratory implementation have not been proved. Accordingly, T_C is called a *carrier*, not an observed metric or stress tensor.

10 Regeneration at every finite stage

10.1 Observation mates

Let $\phi : M_5(\mathbb{C}) \rightarrow \mathbb{C}$ be any state satisfying $\phi(I_5) = 1$, and define

$$E_\phi = \text{id} \otimes \phi. \quad (63)$$

Then

$$E_\phi \circ j_n = \text{id}_{\mathcal{A}_n}. \quad (64)$$

No tracial assumption is required on the refined range. The normalized trace, a point state, and a nontracial full-support state therefore recover the same embedded source record.

For an observer word w represented by a unitary U_w , define the co-moving refinement and readback

$$J_{w,n} = \text{Ad}(U_w) \circ j_n, \quad E_{w,\phi} = E_\phi \circ \text{Ad}(U_w^{-1}). \quad (65)$$

Then $E_{w,\phi} J_{w,n} = \text{id}$.

Let $\mathcal{C}_n$ denote the deterministic composite that maps the source packet at stage n through the source-orbit constructor, geometry, holonomy, and Eq. (58) to its ten-component carrier.

Theorem 10.1 (All-finite-stage carrier naturality). *For every finite stage n , every declared observer word w , and every unital spectator-factor state ϕ ,*

$$\mathcal{C}_{n+1} \circ J_{w,n} = \mathcal{C}_n \quad (66)$$

after the co-moving readback of Eq. (65). The same carrier law and the same coefficients are used at every stage; no stage-dependent refit is permitted or required.

Proof. Equation (64) recovers each refined source component exactly. Conjugating the refinement and composing with the inverse conjugation preserves that identity. The source-orbit, geometry, holonomy, and carrier maps are deterministic functionals of the recovered source packet, hence their composition recovers the same carrier. The exact regression checks cover eight actual source sectors, four source components, and three distinct observation states: 96/96 depth-one readbacks and 96/96 boundary co-moving re-entries. Representative depth-one and depth-two checks pass 6/6 with the same ten-channel carrier hash. $\square$

The theorem is stronger than preservation under a chosen normalized trace: the carrier is independent of which unital spectator state is used on the refined range. It is also weaker than a physical continuum theorem. No limit topology, infinite-mode Hilbert space, or continuum field is inferred from Eq. (66).

11 Main result: a counterexample to primacy

Theorem 11.1 (Finite nonterminal generated-objectivity theorem). *Under the declared source and regularity assumptions, the pointed-braid construction satisfies Definition 2.1. Specifically:*

1. *the relevant coframes and holonomies are outputs of one registered braid-source chain rather than stored target geometry;*
2. *independent internal presentations of A and B are quotiented exactly;*
3. *the joint carrier transforms tensorially under ambient changes of description;*
4. *the joint carrier is unchanged by exchanging A and B but changes when either contribution is collapsed;*
5. *the same source-native carrier law is regenerated at every finite refinement stage without fitting.*

Proof. Item 1 is the exact end-to-end source-orbit geometry result summarized in Eq. (3). Items 2–4 follow from Proposition 9.1 and Theorem 9.2. Item 5 is Theorem 10.1. $\square$

Corollary 11.2 (Failure of Objectivist Primacy as a necessity claim). *If a mathematically admissible countermodel is sufficient to test the universal claim OP, then OP is false.*

Proof. OP says that every stable, shareable, transformation-respecting carrier must be primitive. Theorem 11.1 constructs such a carrier as the nontrivial output of two maintained source standpoints and their intersection law. Therefore the universal quantifier in OP has a counterexample. $\square$

The phrase “counterexample to objectivist primacy” must be read with the antecedent in Corollary 11.2. A philosopher may reject the interpretation of the source prefixes as subjectivities, or deny that a finite formal countermodel constrains metaphysics. Those are legitimate objections. They do not invalidate the mathematics; they contest the bridge from mathematics to ontology. Conversely, the theorem does not license the stronger statement that all mind-independent reality has been disproved.

12 What is created, and what is not

12.1 Objectivity survives, but changes status

The result does not replace objectivity with private experience. The output is public in the relevant mathematical sense: internal presentation choices do not alter it, observer-word refinements admit exact co-moving recovery, and ambient changes act covariantly. What changes is the ontological order. The public structure is the result of a quotient-and-soldering operation, not the assumed stage on which that operation occurs.

This distinguishes generated objectivity from a naive averaging of perspectives. The carrier is not the common scalar part left after every direction is removed. Proposition 8.1 shows that this route is too poor to recover an anisotropic tensor. The construction must retain the directional defect of each standpoint, solder it to a common ambient type, and only then quotient the internal descriptions. Objectivity is achieved by a disciplined relation among standpoints, not by erasing them.

12.2 The role of A , B , and C

Formally, A and B are inequivalent source prefixes with distinct certified holonomies and carriers. Their exchange is a symmetry of the joint output, but they are not identified. C is the joint carrier after internal presentation dependence has been removed. In the proposed ontology, this formal pattern is read as follows:

- A is Absolute Subjectivity, associated with light;
- B is Relative Subjectivity, an impersonal mode associated with time or a clock, not an individual, organism, mind, or conscious observer;
- C is Intersection Subjectivity, associated with space and objective manifestation.

The formal theorem establishes the grammar “distinct A and B generate a nontrivial invariant/covariant C .” It does not establish those physical nouns. Within the ontology, however, B is explicitly pre-personal and pre-biological: relative clocking and ordering are held to belong to the universe before humanity. This is the precise boundary between mathematical formulation and ontological interpretation.

12.3 Creation as regeneration rather than first manufacture

The all-stage law changes the meaning of creation. It is not a single event in which a permanent object is made once and then carried unchanged forever. At each finite stage, a refined source packet can be read back co-movingly and the same law regenerates the carrier. Thus the proposal is compatible with a world whose apparent persistence is repeated structural re-creation.

This does not mean that every stage produces a newly measured nonzero excursion. The fixed-window boundary response is nonzero in all eight certified sectors, but a nonzero boundary excursion of T_C itself

has not yet been proved. The theorem establishes exact regeneration of the carrier law, not a complete physical dynamics of creation events.

13 Relation to nearby programs

Program	Shared concern	Difference from the present claim
Invariantism and structural realism	Objective content is associated with invariants or relational structure.	The present model asks whether the invariant must be primitive. Its invariant/covariant carrier is constructed only after distinct source standpoints contribute directional data.
Relational quantum mechanics	Physical facts or states are relative to physical systems rather than globally absolute.	The present result is not an interpretation of the quantum state. It constructs a specific real Sym_4 carrier and an all-stage regeneration law from a pointed-braid source.
Quantum reference frames	Descriptions transform between physical frames without an external absolute frame.	QRF work supplies precise perspective transformations within quantum kinematics. Here the primary claim concerns the generation of a shared tensor carrier from source intersection, not merely switching among descriptions of a pre-specified system.
QBism	The quantum state is not an objective property; interaction and agent-relative probabilities are central.	“Subjectivity” here is not belief, experience, or conscious agency. The output is an exact inter-presentation tensor, not a personal probability assignment.
Perspectival realism	Reliable knowledge can arise across a plurality of situated perspectives.	Perspectival realism is principally an account of scientific representation and inquiry. The present paper proposes a formal generative ontology and a source-level physical research program.
Braided particle models	Braid topology can encode particle-like structure and interactions [4].	The present use of braiding concerns coherence, source transport, holonomy, and generated objectivity; it does not identify Standard-Model particles with braid words.
Set-theoretic Yang–Baxter models	Permutations of $X \times X$ can satisfy Yang–Baxter independently of this model [2].	The 25-event permutation is not claimed as a new class of Yang–Baxter solution; its fixed spectral-marker provenance and distinction from identity/flip are the checked features. No classification-level inequivalence is established.
Quantum causal histories	Locally finite pre-spacetime causal relations can carry matrix algebras and completely positive evolution [7].	The present source begins with a pointed-braid matrix-algebra tower and tests a particular spectral event action. It has not derived a physical causal continuum or shown an Einstein limit from finite evolution.
Causal-set growth	Discrete causal order and covariance can support sequential growth dynamics [16].	The present address cylinders and braid crossings have different primitives; neither a causal-set embedding nor a physical growth law is proved.
Thermodynamic Einstein route	The Einstein equation can follow from local horizon thermodynamic assumptions [10].	The present route instead assumes a Palatini action and matter/Ward conditions on a selected smooth domain; the source-to-domain convergence is still open.

Program	Shared concern	Difference from the present claim
Metric-affine gauge gravity	Coframe, independent connection, torsion, nonmetricity, hypermomentum, and Noether identities form a first-order gravitational framework [9].	The present work does not assume a continuum affine gauge theory as its microscopic source. It gives finite affine-source structures and actions, but the actual finite-source-to-smooth limit is not yet verified.
Earlier Subjectivity Intersection work	Pointed braids provide all-orders intersection consistency and compositional objectivity [20].	The present version retains the v0.6.1 finite geometry and action chain, then adds fixed-model MMR2 selection, source-derived 3/5, and an exact normalized Lorentz event carrier. The source-to-continuum and SDPC obligations remain explicit.

The novelty claim is therefore conjunctive. Neither relationality, perspective dependence, invariance, braid groups, nor process-like ontology is individually new. The contribution is the exact chain

$$\begin{aligned}
& \text{distinct braid standpoints} \longrightarrow \text{target-free } 1 + 3 \text{ carrier} \longrightarrow \text{source-only holonomy} \\
& \longrightarrow \text{direction-preserving soldering} \longrightarrow \text{joint covariant carrier} \\
& \longrightarrow \text{finite metric-affine dynamics} \longrightarrow \text{curvature and defect actions} \\
& \longrightarrow \text{finite spectral events and a selected smooth-domain matter action} \\
& \xRightarrow{\text{source limit}+\text{SDPC}} \text{conditional Einstein equation,}
\end{aligned} \tag{67}$$

together with an explicit philosophical argument showing what this chain does and does not refute.

14 Operational content and falsification

14.1 A reproducible mathematical readout

The current operationalization is a deterministic computational readout, not a laboratory apparatus. Given the registered source packet, it performs the following steps:

1. generate the opposite source slice using the frozen signed-braid action;
2. reconstruct the affine geometry and the two source-prefix coframes;
3. compute the source-only face holonomies;
4. form D_p , M_p , T_p , and T_C by Eqs. (55)–(58);
5. output the ten independent entries of $T_C \in \text{Sym}_4$;
6. verify internal-gauge invariance, prefix-exchange invariance, ambient congruence, and the three collapse controls;
7. refine by j_n , apply a selected unital spectator state, and verify Eq. (66) without fitting.
8. construct the source graph, verify finite propagation and the boundary-event response, and evaluate the all-stage Ward kernel in Eq. (44);
9. build the continuous certified face representatives, decompose the regular source by Eq. (46), and verify the exceptional positive quotient channels;
10. fix the UB476 response before geometry, apply the three declared local-Einstein bridges, and verify the exact Riemann, Bianchi, Ricci, Einstein, and metric-two-jet identities in Eq. (14);
11. evaluate the finite action in Eq. (50), its coframe, connection, and metric variations, and the local-frame identity in Eq. (51);
12. verify the source-library balance no-go without replacing it by a fitted negative Euler residual;
13. evaluate the independent defect action in Eq. (53), its 832 exact variation checks, connector ranks, covariance tests, and Noether identities.

14. generate the one-site marker's exact spectral event projectors, replay the actual 25-event braid permutation and its refinement/readback controls, and distinguish this formal event law from an empirically calibrated physical event;
15. evaluate the original ten F_A , the local doubled matter action and Eq. (19), then check Eq. (21) both with RML and against the bare, reverse-link, and four-address controls;
16. replay the complete five-partition fixed-packet test, verify that only the full corner preserves the eleven-operator packet, and recover the normalized source cup 3/5 without target fitting;
17. reconstruct Q_4 , J_{ray} , and K_{evt} from the actual event transition, verify the exact $(3+, 1-)$ signature and unique S_4 -equivariant normalization, and replay the identity, flip, orientation, and closed-time-path controls.

This is already a nontrivial readout because it returns directional tensor data that scalar invariants provably cannot supply in the declared linear equivariant class. It is not yet an empirical observable because no instrument model, units, noise model, or identification with measured geometry has been given.

14.2 Formal falsifiers

The present theorem package would fail if any of the following occurred under exact replay:

- the target geometry entered the source constructor before comparison;
- either internal GL_3 presentation changed T_C ;
- A/B exchange changed the joint carrier;
- a one-sided collapse reproduced the joint carrier;
- the ambient carrier failed the congruence law;
- the all-stage readback required a tracial spectator state or a stage-dependent coefficient;
- the source-graph support propagated faster than the certified graph bound;
- the $3 + 1 + 5$ source product failed to reconstruct the registered geometric source;
- the source-first metric two-jet failed any algebraic Riemann symmetry, Bianchi, contraction, Lorentz-signature, or exact Einstein check under its declared bridges;
- the analytic action variations failed their exact central-difference checks;
- the local-frame Noether identity failed with all required Euler slots included;
- the covariant-defect action accessed the curvature residual or an Einstein target, failed to change off shell, or lost its stated connector rank;
- the exact marker spectrum, 25-event braid transition, five-child cylinder identity, or co-moving readback failed on a signed sector;
- any original F_A fell outside the declared event/energy readout span, or the full- W integration failed on a tilted source state;
- the RML identity or its historical 3/5 cup failed while the stated corner normalization and source ranks held;
- a non-full partition preserved the complete frozen event-conditioned packet, the selected normalized source cup failed to equal 3/5, or either result required comparison with the Einstein target;
- the actual-event Q_4 lost rank or covariance, K_{evt} lost its exact Lorentz signature, or the declared naturality class admitted a second positive normalizer;
- any registered exact regression ceased to reproduce its frozen source record.

14.3 Physical discriminators still required

To test the Creation Principle as a claim about nature, at least one bridge must produce an empirical discriminator. Four routes are open:

1. identify T_C with, or derive a calibrated map to, an operational metric or stress response;

2. test the frozen coefficient-one sum of the curvature and defect actions, then show that any resulting stationary equations possess a controlled continuum and low-energy sector;
3. derive a boundary/re-entry signal that standard quantum-reference-frame or effective-field descriptions do not predict;
4. show that a known physical structure is recovered with fewer independent primitive assumptions and with a novel constraint among measurable channels.

The next internal test is no longer another RML or coefficient scan. The shortest remaining route is to construct a first-order phase-space or Legendre parent action from the marked Braid generator and ask, in one calculation, whether its principal symbol selects J_{ray} on the same event-gradient field, its first jet reproduces the already fixed full-corner matter responses, and its diagonal symmetry yields the stress Ward identity. This is the decisive SDPC test. Actual source-to-smooth convergence and convergence of first variations remain a separate continuum gate.

“Measuring what was previously unmeasurable” would count as a genuine contribution if the new readout has an independently specified instrument, calibration, error model, and discriminating result. A renamed internal variable would not.

15 Objections and boundaries

15.1 Where is quantum theory?

The source is quantum in a specific algebraic sense: it uses complex noncommutative matrix algebras, states, unitary pointed-braid transports, tensor refinement, and adjoint observer actions. Those ingredients are enough to make the classical-output result nontrivial, but they are current inputs. The paper does not derive complex amplitudes, the Born rule, Hilbert-space composition, unitary time evolution, or quantum-field locality from a bare topological braid group. Accordingly, “a route to quantum structure” names the wider Braid-first programme. The exact result of this paper begins at a quantum-algebraic Braid source and derives the declared classical carrier and finite gravity dynamics from it.

15.2 Where is spacetime, and does the ambient four-space smuggle objectivity back in?

Version 0.5 already removed the numerical and ambient-carrier forms of this objection at the mathematical layer. The finite source produces $Q_{\text{obs}} = V_4$. Full affine-observer symmetry, an order unit, and point separation force the complete readout and hence the target-free split $1 \oplus 3$. Observer Uniform Cauchy Closure completes that carrier to $W = \mathbb{R}^{V_4}$ without adding a direction; $\text{Aff}(W)$ is then its own smooth four-dimensional affine carrier. The source path is not renamed as physical three-space, and the four metric-response controls are not renamed as region coordinates.

Version 0.6 added a checked finite event algebra and crossing/readback rule. The v0.6.1 code audit corrects the stronger conclusion drawn from it: BGCE138 constructs a smooth completion and compatible samples, but does not show that actual finite source fields converge to that completion. Post-v0.6.1 results retire the independent finite event-algebra correspondence inside the fixed model, but they do not prove the reverse source-to-smooth arrow. Thus formal CGR as a complete source-to-continuum statement remains open. Physical identification with measured events, dimensionful calibration, and continuum first variation for nature are additional requirements.

15.3 Is the carrier unique?

No global uniqueness theorem is claimed. Equations (56)–(58) are a natural, exact, source-native solution that obeys the declared symmetries and controls. Other nonlinear or higher-order carriers may exist. The

counterexample to OP needs existence, not uniqueness. A stronger physical theory would benefit from deriving the carrier from an action, universality principle, minimality theorem, or experimental constraint.

15.4 Is a finite model enough?

It is enough to refute a universal mathematical necessity claim if the model is admitted. It is not enough to establish a fundamental theory of our universe. The nonterminal tower prevents the result from being only one isolated finite face, while the source graph, spectral events, and continuous certified cells add mathematical notions of locality and curvature. They do not by themselves produce an empirically identified global continuum spacetime. The historical CGR/MMR theorem supplies a conditional local infrared Einstein equation. The v0.7 fixed-packet theorem removes MMR2 as an independent routing postulate inside its declared class and the event moment supplies an exact Lorentz carrier, but source-to-continuum convergence and SDPC remain independent premises. Unitary quantum-field evolution, renormalized stress, global extension, and empirical scale identification remain separate tasks.

15.5 Where is gravity?

The construction contains more than prerequisites for gravity. Equations (8) and (9) give an exact source-native coframe update and covariant curvature closure, while Eq. (10) gives the corresponding finite metric-affine update. Rank-changing geometry requires the source term, and every nonmetricity, source-linear, and source-quadratic contribution survives ablation. The same packet extends without refitting through all finite stages. The source graph supplies finite locality and a Ward law; continuous certified faces carry nonzero connection curvature; Eq. (50) supplies an exact first-order curvature action with Eq. (51) as its local-frame identity; and Eq. (53) supplies an independent off-shell variational connector. In this exact fixed-orbit sense, an action-bearing finite Braid metric-affine gravity candidate has been constructed.

The relation to Einstein gravity has two levels. First, the independently fixed source generates a local Lorentz metric two-jet satisfying Eq. (14) at one point under three explicit bridges. Second, the off-shell action (16) yields Eq. (18) on a neighborhood under the stated continuum and matter conditions. The second is a genuine *conditional* Palatini variational result, not a derived finite-source infrared limit. The v0.7 fixed-packet theorem selects the full matter corner and its 3/5 relative normalization in the declared class, while the actual event moment and source-null grading give an exact Lorentz carrier. What remains open is not manual selection of that coefficient: it is actual source-to-smooth convergence and SDPC—action-level grading, physical coframe/stress pullback, nontrivial physical-section dynamics, and the Ward identity from the same parent action. Finite combined stationarity, physical units, quantized continuum gravity, and empirical identification also remain open. The noninvolutive four-flow result excludes one first-class route but does not negate the single local GKSL evolution. This is a Braid-based quantum-gravity candidate with exact finite mathematical results and a conditional Einstein construction, not completed quantum gravity or source-only general relativity.

15.6 Is “subjectivity” only a relabeling?

It would be a relabeling if the formal roles made no difference. Here the roles are constrained: two prefixes must remain distinct; their internal presentation can be quotiented but their contributions cannot be collapsed; and their intersection produces a carrier with a verified transformation law. This is more than terminology. Nevertheless, calling these formal roles “subjectivities” is an interpretive move. The mathematics fixes the relational structure; ontology assigns its meaning.

16 The Braid-first Creation Principle as a research programme

The present work suggests a sequence of progressively stronger claims:

1. **Quantum-origin adequacy:** quantum operational structure is derived from less assumed Braid data rather than placed in the source algebra. This remains open.
2. **Formal generation:** distinct Braid standpoints generate exact classical objective structure. This paper establishes that claim for the declared quantum-algebraic source.
3. **Finite dynamical adequacy:** the source law generates exact coframe and metric-affine updates, a finite local net, curvature-bearing continuous cells, a curvature action, and an independent covariant-defect action with exact local-frame Noether identities. This paper records that result.
4. **Conditional pointwise Einstein adequacy:** under three explicit bridges, the independently fixed source generates a Lorentz metric two-jet satisfying $G_{\mu\nu} = (3/5)\Theta_{\mu\nu}$ exactly at one point. This paper establishes that bounded arrival result.
5. **Conditional continuum Einstein adequacy:** the v0.5 theorem yields $G_{\mu\nu} = (3/5)T_{\mu\nu}$ on a neighborhood under its continuum realization assumptions. The finite event part of CGR is checked, a formal MMR1 action is constructed on a selected smooth domain, and v0.7 closes MMR2 with source-derived 3/5 inside the complete declared partition class. Actual finite-source convergence and SDPC remain open.
6. **Stationary-field adequacy:** independently frozen source functionals balance every coframe, connection, and metric Euler slot without target fitting. The required connector now exists, but exact stationarity of the coefficient-one combined action has not yet been tested.
7. **Physical-sector adequacy:** the generated carrier, action, and affine-source sectors map to calibrated metric, stress-energy, hypermomentum, or other operational fields. This remains open.
8. **Unconditional continuum and low-energy adequacy:** the source law itself generates the bridge assumptions, a global local-causal spacetime dynamics, and established low-energy gravity. The conditional local equation is obtained here, but this stronger source-only claim remains open.
9. **Empirical selection:** a measurement distinguishes the Creation Principle from lower-commitment relational or effective descriptions. This remains open.
10. **Ontological adoption:** the Creation Principle is accepted as the best account of what reality is. This would require the preceding physical and empirical work plus philosophical argument; it cannot be obtained from the finite certificates alone.

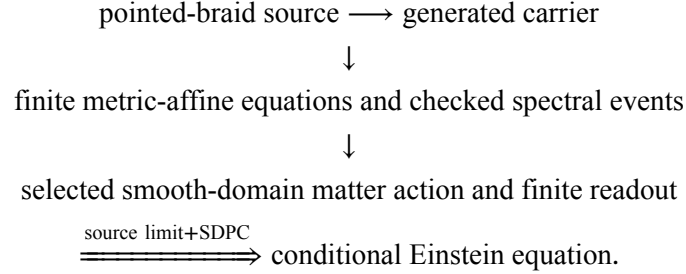
This hierarchy is not a retreat from the ontological thesis. It states what would make that thesis scientifically serious. The first level changes the dialectical situation: objectivist primacy can no longer be defended merely by asserting that no exact alternative can produce stable public structure. The remaining dispute concerns physical realization, explanatory economy, and evidence.

17 Conclusion

Subjectivity-Intersection Braid Quantum Gravity asks whether quantum structure, spacetime, and gravity can be recovered from one standpoint-bearing braid law. Its exact finite core is now substantial. A quantum-algebraic pointed-braid source with two distinct standpoints generates nontrivial directional holonomies. If these are reduced too early to scalar invariants, an exact rank obstruction prevents reconstruction of a full anisotropic tensor. If instead each holonomy defect is soldered through its source-generated coframe, internal presentation dependence disappears while ambient direction survives. Their sum yields a real rational Sym_4 carrier that is exchange-invariant, covariant, nonzero, collapse-separated, and reproducible by one law at every finite stage of a nonterminal tower.

The same source lineage produces a target-free $1 + 3$ carrier, finite coframe and metric-affine equations, a source-local graph and GKSL evolution, certified faces with nonzero connection curvature, a $3 + 1 + 5$ post-Riemannian source decomposition, and curvature and defect actions with local-frame identities. Version

0.6 added a checked finite spectral event algebra, a selected smooth-domain doubled matter action, finite ten-operator readout, and RML-conditional full- W routing. Version 0.7 then shows that the complete frozen event-conditioned packet selects the full corner among all five source-atom partitions, making the relative 3/5 normalization source-derived in that declared class; the actual event transition and source-null grading also yield an exactly normalized Lorentz carrier. The pointwise result constructs an Einstein-compatible Lorentz metric two-jet under explicit bridges. Under actual source-to-smooth convergence and SDPC, the Palatini action (16) yields Eq. (18). The corrected dependency chain is



This enlarged chain strengthens rather than displaces the refutation of objectivist primacy. The counterexample no longer generates only a stable shared tensor. It generates finite local geometry, curvature, variational response, and, conditionally, an Einstein field law on continuous spacetime. Objectivity survives as covariance, reproducibility, and inter-presentation stability; what fails is the universal claim that such objective structure must be ontologically prior to every standpoint relation. This is a constructive refutation of objectivist *primacy*, not a denial of objective knowledge and not an empirical disproof of mind-independent reality.

The quantum-gravity claim remains bounded. Full formal CGR retirement is not established because the actual source-to-smooth reverse arrow remains unproved. MMR2 is no longer an independent postulate inside the declared fixed event-conditioned partition class, and 3/5 is no longer manually set there; neither result is yet a completed physical matter law. SDPC remains a non-Braid-only completion assumption, and the continuum gravitational field has not been quantized. The current temporal rank-four response distribution is noninvolutive, although the combined local GKSL evolution exists. Ultraviolet completion, physical scale, and a discriminating natural-data prediction remain open. The Creation Principle makes the stronger ontological proposal that objective manifestation is continually generated when distinct subjectivities intersect without becoming identical. The present paper supplies that proposal with a target-free $1 + 3$ mathematical carrier, an exact finite foundation, a checked finite event algebra, a source-derived Lorentz event carrier, and a conditional Einstein construction; its physical truth remains open to assumption removal and empirical selection.

Data, code, and provenance

The exact certificates used in this paper are stored in the SIEL Research Agent repository. The v0.2 carrier and finite-gravity chain is preserved at commit `3d1cccc86`. The post-v0.2 locality, conditional local-Einstein, continuous-cell, metric-affine, action, Noether, source-balance, and covariant-defect connector lineage is preserved through commit `27dd70da0`. The first continuum derivation and its historical four-law audit are preserved in BGCE038. BGCE075–BGCE127 preserve the temporal rank, nonclosure, action, normalization, KMS, inductive-limit evolution, and quasilocal-locality results. BGCE128 removes the primitive numerical and ambient four-carrier input at the mathematical layer; BGCE129 records the then-current CGR/MMR dependency packet. These v0.5 records were preserved through commit `75d2c2691db515e35ef8bdec82047b0c1d56306f`. Version 0.6 cited BGCE139 for formal CGR retirement; this v0.6.1 code audit withdraws that inference while retaining its finite event check. BGCE138 constructs a compatible smooth completion, BGCE099 states the UTC-conditional continuum

action, BGCE142–143 construct a selected-domain doubled matter action, BGCE185/187 support finite readout and analytic reconstruction, BGCE197 verifies the historical RML-conditional algebraic routing, and BGCE199, BGCE205, and BGCE206 delimit specified implementation classes. Post-v0.6.1 BGCE234–BGCE260 preserve the fixed-packet MMR2 selection, non-manual 3/5 routing, event moment, null-ray grading, Lorentz carrier, naturality, and SDPC/no-go audits. The inspected post-v0.6.1 source snapshot is SIEL Research Agent commit [eb61e91a0](#). The repository is [SIEL-Research/SIEL-Research-Agent](#). The gate records contain protocols, raw outputs or analytical claims as appropriate, verifiers for some finite endpoints, and audit manifests. Their coverage is heterogeneous: a stored PASS field is not a proof of every prose-level implication, and several continuum/readout claims have no executable test of their full scope. This manuscript does not bundle the complete source code or source data; independent reproduction requires retrieval of the pinned repository paths and commits. The reported results are exploratory theoretical and computational results; this code inspection is not independent numerical replication, peer review, or empirical validation.

Declaration on computational assistance

OpenAI Codex was used under the author’s direction for mathematical ideation, exact-check scripting, audit organization, literature triage, and drafting. The author retains responsibility for definitions, interpretation, verification, citations, and any decision to release or submit the work.

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